Degree-limited actual whole-chart factors

14 November 2026; local29 statements; data PCC-20261114-36a2d2

Local construction. Accepted draft004/build004 elaborates29 statements,exit0/no warnings. This separate draft is OUTSIDE Compact39's shared audit and fresh replay. Five central printed dependencies contain only standard foundations. All observations are unsigned [pcc:unverified].

Actual scope. Use the same completed auxiliary W, actual bounded ports $x, y < 2^a$, natural multiplicities, and nonzero signed-grid source node t with $v_2(t) = i < u \le a$. Matrices explicitly mask noncrossing entries to zero. The actual fixed negative-pole diagonal is c, with $w = v_2(c)$. No positive multiplicity or nonzero local-first-jet premise is added.

Combined grading. Coefficient H^b of the actual bivariate geometric approximation has Z-grade $b-1$. Dividing by 2^b gives grade -1 of the actual constructed right factor $V_{(r,b),y}(Z)$:

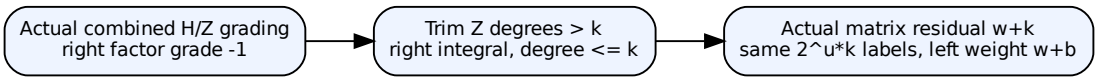
$$v_2([Z^n]V_{(r,b),y}) \ge n-1,$$

with zero coefficients admitted at every weight. The grade is derived from actual denominator slopes and the existing actual normalized chart, rather than supplied as a good-factor hypothesis [pcc:unverified].

Define $V^{\le k}$ by truncating each actual right polynomial below degree $k+1$. The coefficient bound gives content-k error. The finite matrix product error is proved using actual left integrality and sums, then scaled by c. The resulting profile is

$$\operatorname{natdeg} V^{\le k}_{(r,b),y} \le k, \qquad \operatorname{Content}_0(V^{\le k}_{(r,b),y}),$$
$$\operatorname{Content}_{w+b}(\widehat{U}_{x,(r,b)}), \qquad \operatorname{Content}_{w+k}(W^{\mathrm{cross}}_{xy} - (\widehat{U}V^{\le k})_{xy}),$$
$$\#\{(r,b)\} = 2^u k.$$

Every label remains charged; $k = 0$ is included. Scaled-left integrality retains $w \ge 0$. No full-matrix or origin rank claim [pcc:unverified].



Evidence and failures. Exact accepted source/log/exit are pinned in local-validation-002.json; prior21 data is PCC-20261114-20c45d. All four drafts/raw builds are retained. Failures include a reserved-lambda variable, an omitted unfolding and ambiguous weightedLeft. Accepted004 has no `sorryAx`; no linter or acceptance condition was disabled [pcc:unverified].

Next discrimination. Independent promotion/audit/replay remains pending. Degree and labels do not price coefficient construction, bit sizes, entry or creation precision, original mode density, recovery, recursion, actual elliptic relation/public finder/extraction/retry/expected bit cost. Read exact Mixed/Graded, actual Compact and pinned PowerSeries truncation APIs before this derivation; KB unavailable as journaled. Standard algebra is reused without a novelty or speed claim. No native crypto, solver, timing, paid or cloud experiment; all parent pins remain exact.